MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION COMMUNICATION AND TECHNOLOGY

ICT 4703: NETWORK ADMINISTRATION AND INFORMATION SECURITY

END OF SEMESTER EXAMINATION

DATE: DECEMBER, 2015 **TIME ALLOWED: 3 HOURS**

INSTRUCTIONS

i. This paper contains **Three** pages. **Please check**.
ii. There are **Four** questions in this paper. **Answer ALL**.
iii. Start answering each question on a fresh page.
iv. Marks for each question are indicated.
v. Write your answers clearly.

Question One

a) Describe the Security Trinity. **[6 Marks]**

b) Secret key encryption is said to be symmetrical. Why? **[2 Marks]**

c) Encryption is not security. Discuss. **[7 Marks]**

Question Two

a) Describe any **three** classes of network attacks that you know. **[6 Marks]**

b) Explain subnetting. **[2 Marks]**

c)

i. State the **two** types of subnetting. **[2 Marks]**

ii. Distinguish between the **two** types of subnetting mentioned in question two **c (i)** above. **[4 Marks]**

iii. Explain in detail the advantage in using each of the two types subnetting mentioned in Question two **c (i)** above. **[6 Marks]**

iv. Explain in detail the disadvantage in using each of the two types subnetting mentioned in Question two **c (i)** above. **[6 Marks]**

Question Three

a) Distinguish between the following:

i. Enable secret and enable password. **[2 Marks]**

ii. Accountability and attack. **[2 Marks]**

b) Two of the disadvantages of symmetric encryption are that it does not provide for authentication and nonrepudiation. Explain how this is so. **[4 Marks]**

c) Given a Cisco Router, provide the **commands** you would write to configure the router, **the mode** in which the command would apply and how to go into that particular mode in doing the following:

i. How to set an encrypted password. **[3 Marks]**

ii. How to get into serial interface 0/0/0. **[3 Marks]**

iii. How to give the interface mentioned in question three **c** (i) above an IP address. **[3 Marks]**

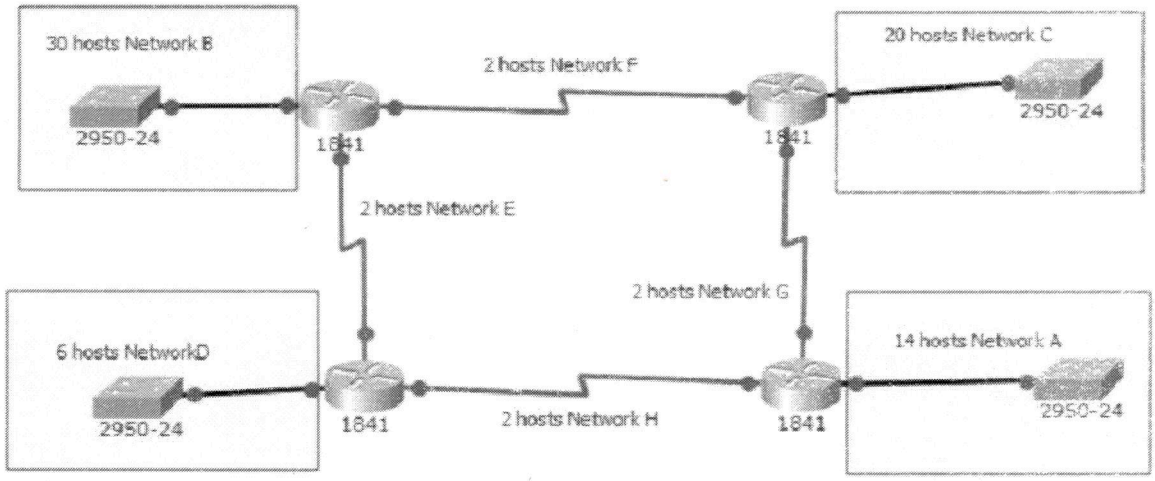
iv. How to bring up the interface mentioned in question three c (i) above. **[3 Marks]**

v. How to give an interface a description. **[3 Marks]**

vi. How to make the router display a message of the day. **[3 Marks]**

vii. How to supply a name to a router so that it is easily identified. **[3 Marks]**

d) Use the diagram below to answer the questions that follow:



i. Variable length subnet masks (VLSM) networking is to take one network and create many networks using subnet masks of different lengths on different types of network designs. Given the network address 192.168.10.0 and using VLSM, do the following:

1. Create **four** subnets that meet the requirements of the design as depicted by the diagram above and for each LAN state its subnet. **[4 Marks]**
2. Using the four subnets you have arrived at, complete the table below:

Subnet	Mask	Subnets	Hosts	Block

[10 Marks]

Question Four

a) Explain cryptanalysis. **[2 Marks]**

b) Describe brute force and social engineering. **[4 Marks]**

c) Most vulnerabilities can usually be traced back to **three** sources. Explain these sources.

[6 Marks]

d) Information security is also about procedures and policies that protect information from accidents, incompetence, and natural disasters. Explain three issues that such policies and procedures address. **[6 Marks]**

e) Alice and Bob communicate by way of sending each other text messages which are encrypted using a key. The key used is: **$C= ((k_2)^2-6)(P)$**.

Hint: **k_2** means two steps forward. **C** means cypher text and **P** means plain text. Decrypt the message below that was exchanged between Alice and Bob.

Y APCYRGTC KYL GQ KMRGTYRCB ZW RFC BCQGPC RM YAFGCTC LMR ZW RFC BCQGPC RM ZCYR MRFCPQ **[10 Marks]**

END OF QUESTION PAPER